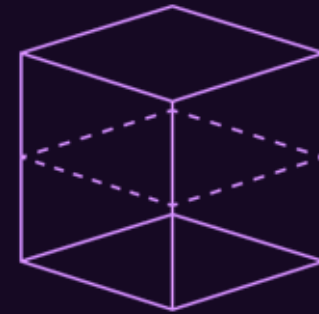




Refactoring The Enterprise



Ash Hunt | VP of Strategy, Cyera

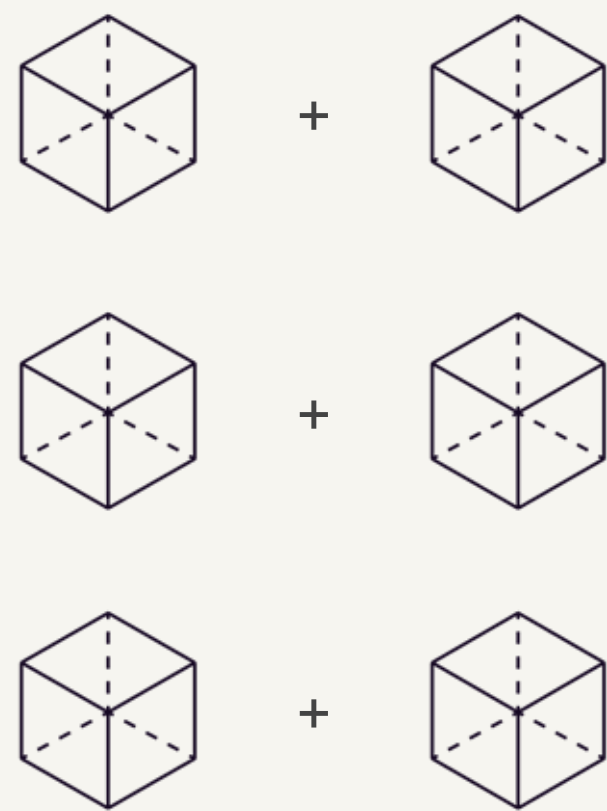
“In the past the man has been first; in the future the system must be first”

Frederick Winslow Taylor (1856–1915)

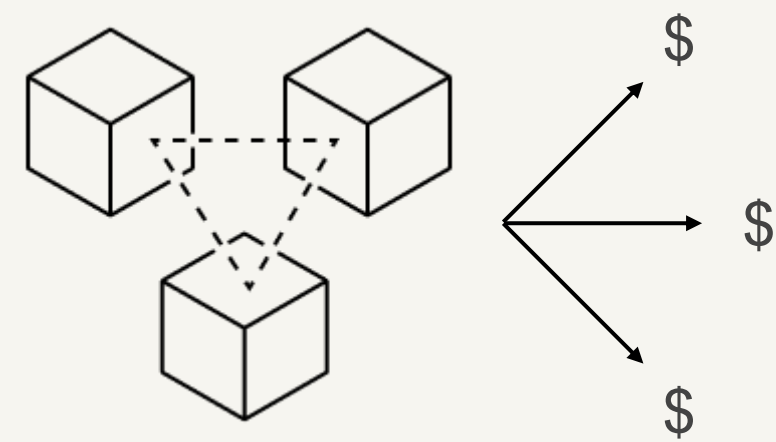


First Principles

What is the basic purpose of an enterprise?



Build

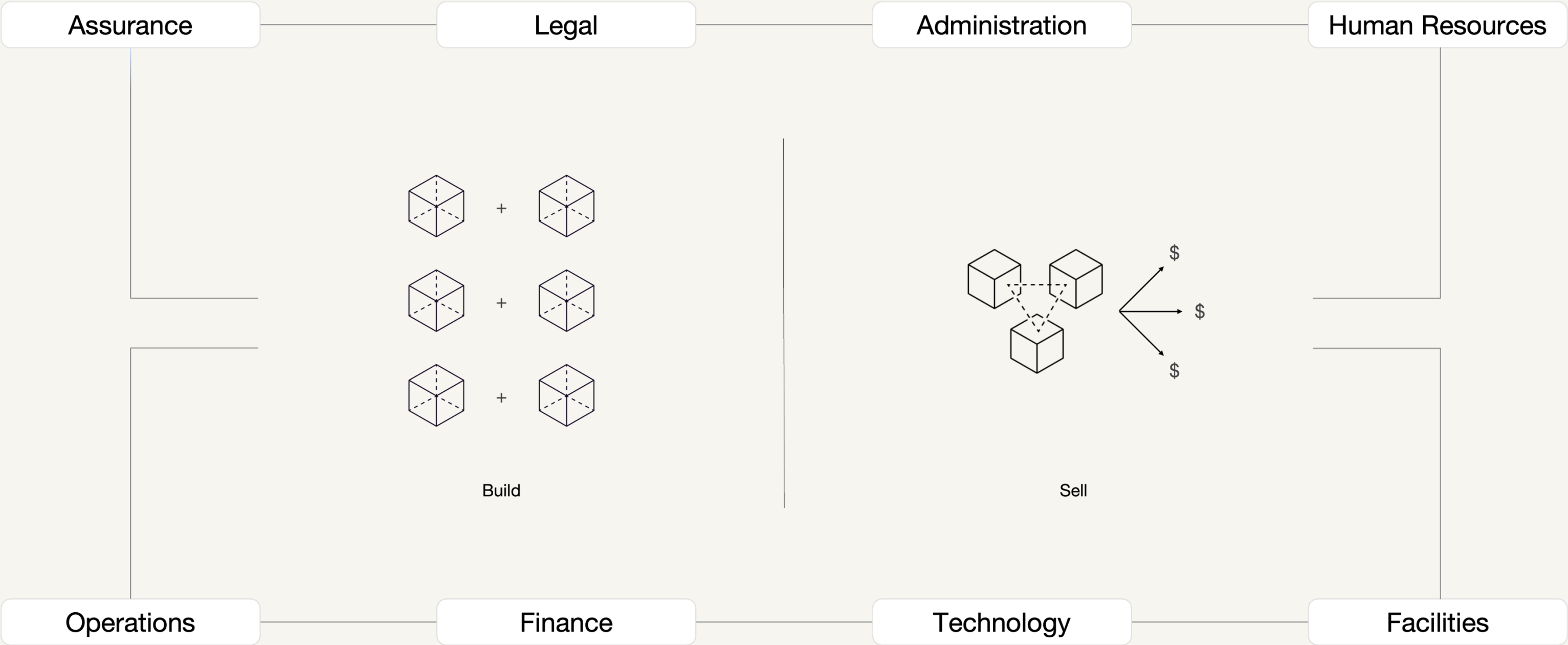


Sell



Scaffolding

How did we optimise the enterprise?



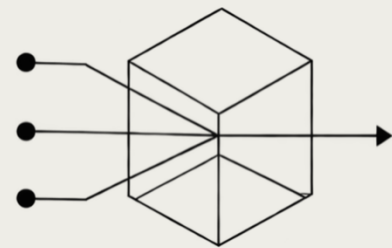
Two Ways The Scaffolding Collapses

How is the enterprise transforming?

System 1 Models

Fast, low-cost, high-volume, structured and confidence-scored decisions within a predefined answer space, without generating free-form text

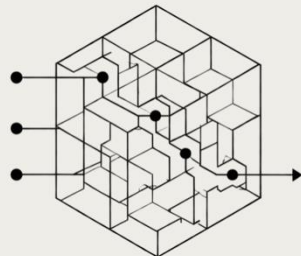
- TypeSafe AI, AWS Strands Labs
- Fast, typed, bounded decisions
- Scores, labels, probabilities
- Reinforcement Learning for Calibrated Decisions (RLCD)
- Lower cost/latency



System 2 Models

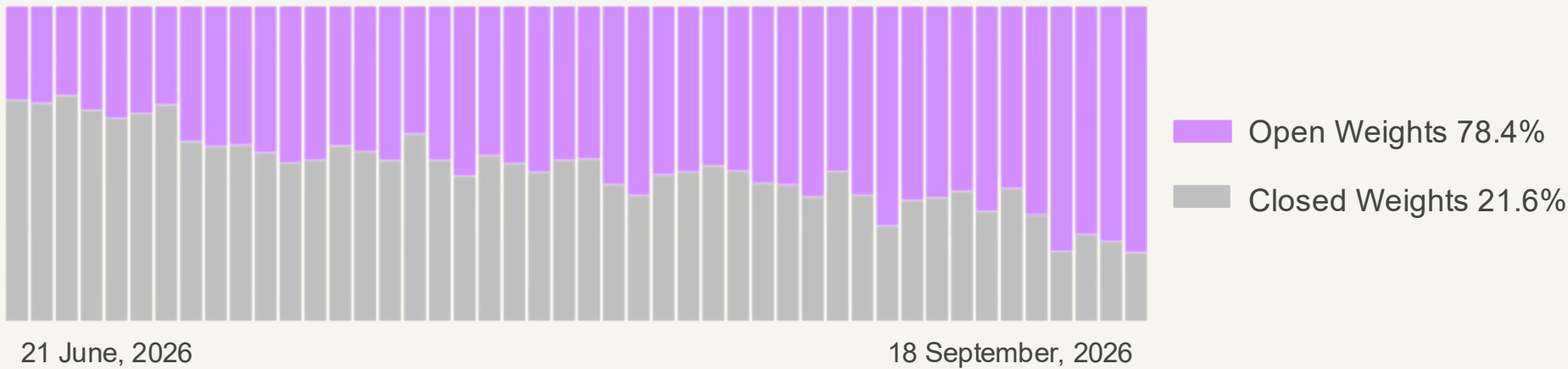
Slow, compute-intensive, multi-step reasoning to interpret ambiguity, plan, verify intermediate work, and generate open-ended answers

- OAI, Google, Anthropic
- Slow, deliberate reasoning
- Free-form text, code, prose
- Reinforcement Learning from Human Feedback (RLHF)
- High cost/increased latency



Open vs. Closed Token Volume

Share of AI Gateway token volume on models that publish their weights for download, versus everything else*



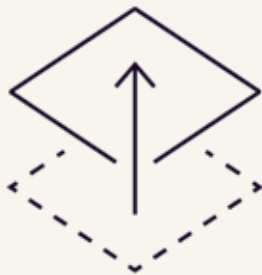
*vercel.com/ai-gateway



Dismantle & Rebuild

How is the enterprise transforming?

Productivity grew about 1.5% a year from 1973 to 1995, then about 3.3% a year from 1995 to 2003, once work was reorganised around the machines*



e.g. ITSM

SYSTEM 1

Classifies ticket intent, detects probable CI, scores urgency, predicts resolver group, rejects malformed requests, identifies likely duplicate incidents in milliseconds

SYSTEM 2

Investigates a cross-domain incident: correlates service topology, recent changes, logs, traces, prior postmortems, vendor advisories, and proposes a remediation plan with uncertainty and evidence

e.g. FinOps

SYSTEM 1

Categorises spend, detects idle or abnormal consumption, allocates shared costs, flags reservation coverage gaps, and forecasts near-term budget variance

SYSTEM 2

Explains cost causality across architectural choices, compares commitment strategies, evaluates migration economics, recommends trade-offs among latency, sovereignty, and cost

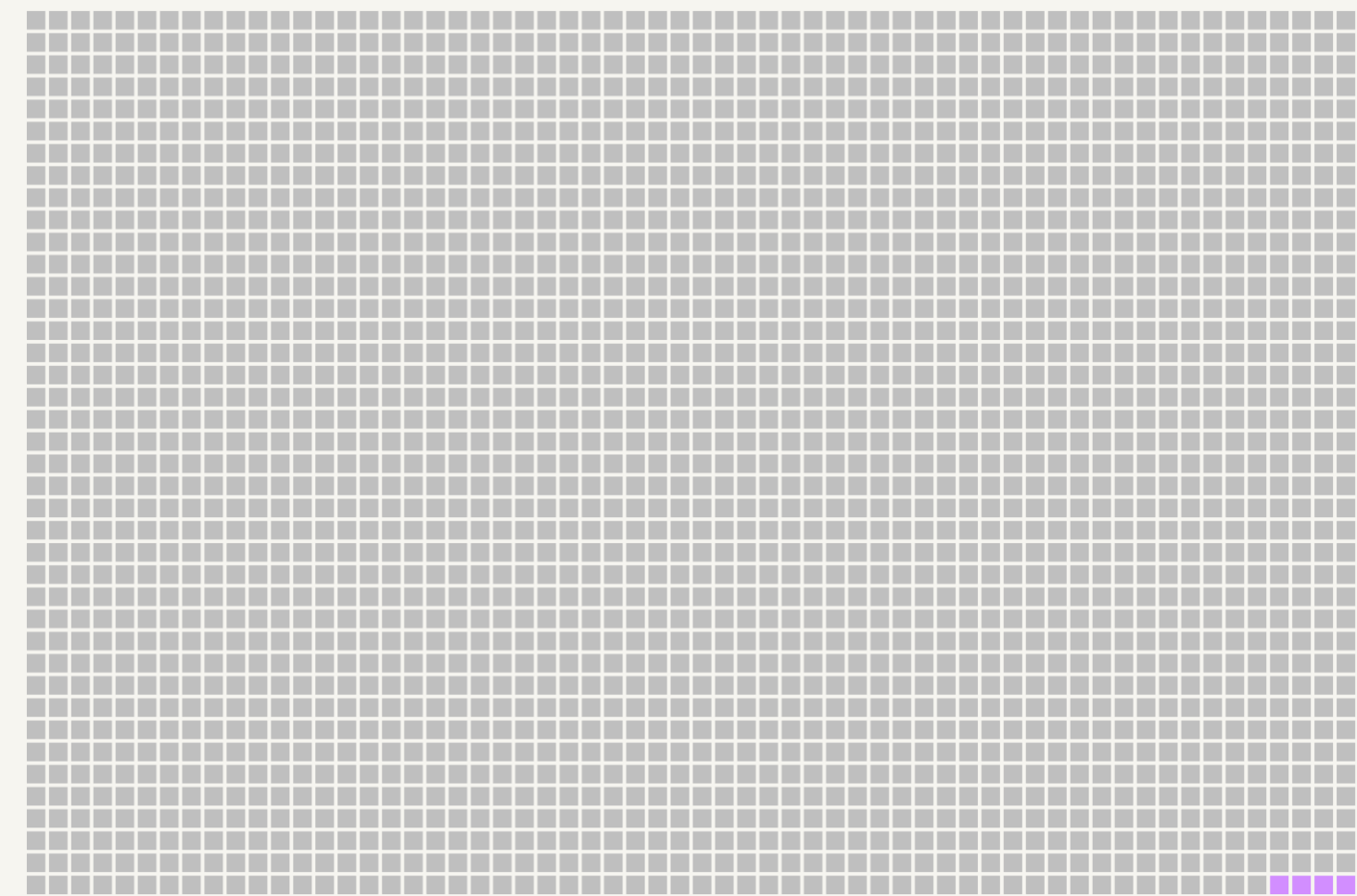
*Roger W. Ferguson Jr., “Lessons from Past Productivity Booms,” Federal Reserve Board, 7 July 2004



Workforce Spend Return

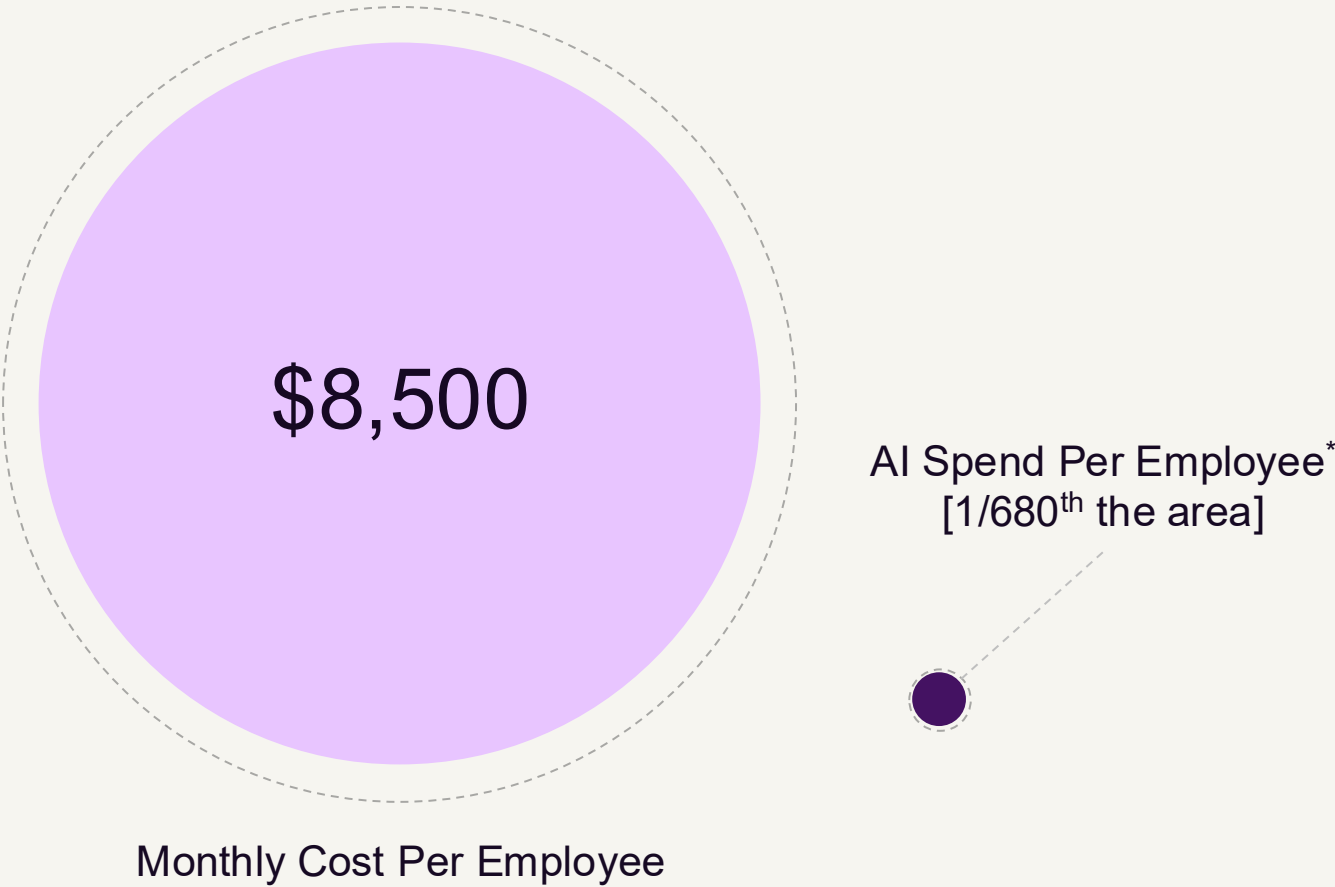
How is AI investment per headcount justified?

The 40 Hour Work Week—2,400 minutes



~3.5 min

Purple squares = break even
0.147% of the working week



Ratio Bar

AI = 0.15c in every \$1



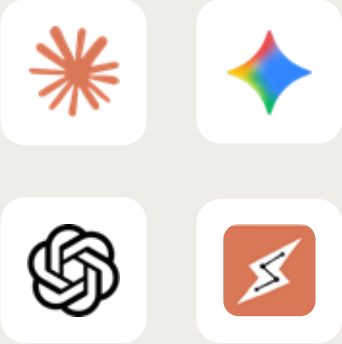
ROI if AI saves 1hr/week

~17 x payback

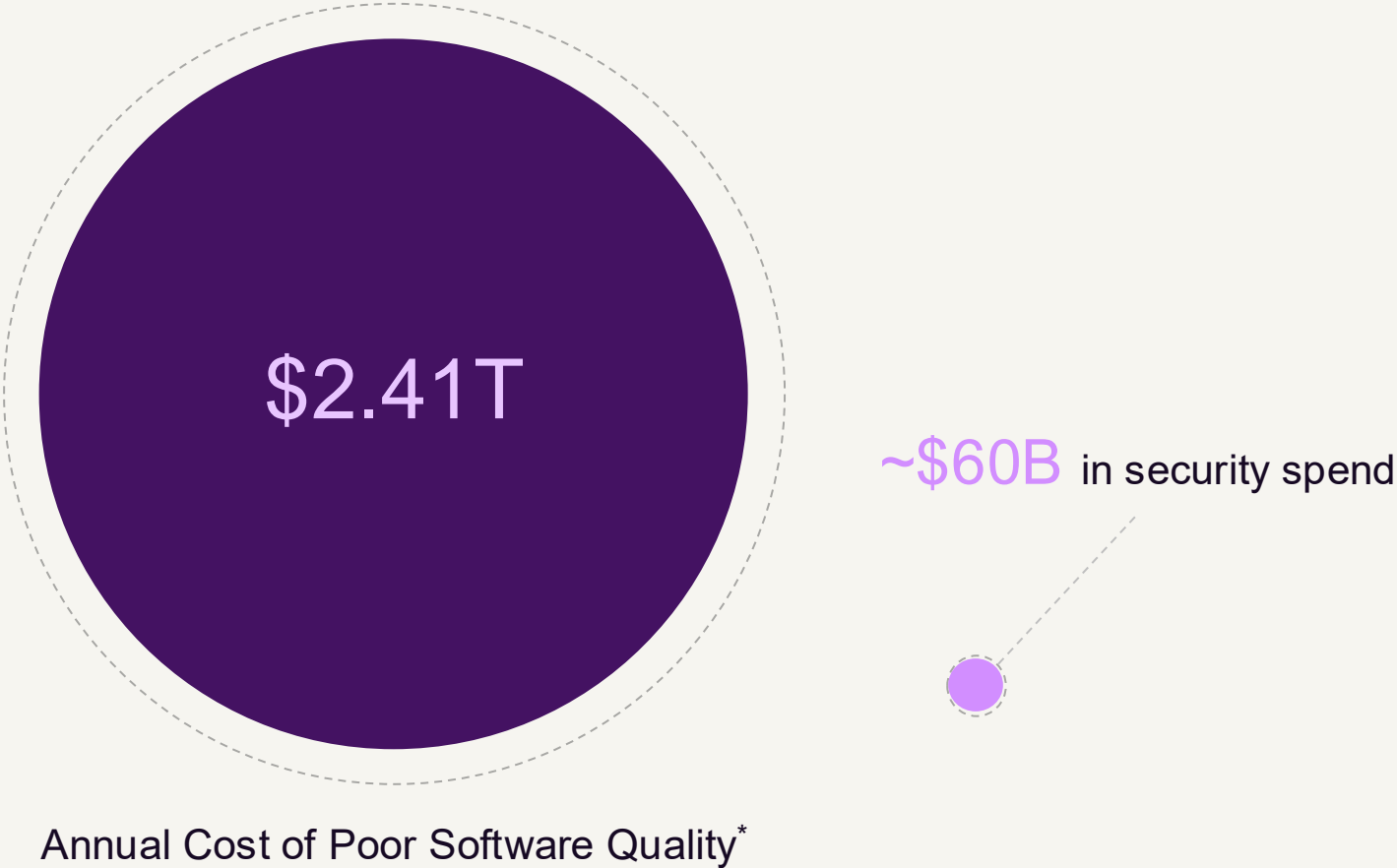


Price Of Non-Conformance

What are the consequences of failing to do things right the first time?



As AI enables transforms the stack, it in turn provides rich context and sweeping transparency to the Price of Non-Conformance



Price of Non-Conformance	Non-Conformance	Explanation
An AI support agent exposes one customer’s account information to another through an overly broad data-retrieval permission	The agent failed the requirement to enforce tenant isolation and least-privilege data access	The organisation incurs forensic investigation, access remediation, customer notification, legal review, support effort, and possibly contractual or regulatory costs. None would be needed if access control had conformed from the start
A customer-service agent retrieves context from the wrong customer identity graph following an entity-resolution error, then uses that context to offer an inappropriate retention deal or disclose account-specific information to the wrong person	The workflow failed requirements for entity matching, tenant/customer isolation, confidence thresholds, and escalation to a human when identity or context is uncertain	Cost includes incident triage, record correction, agent/workflow redesign, complaint handling, goodwill payments, privacy assessment, notification, plus churn. The cost would not have existed had the workflow correctly conformed to identity, data-quality, and privacy controls

*Consortium for Information & Software Quality (CISQ)’s The Cost of Poor Software Quality in the US: A 2022 Report

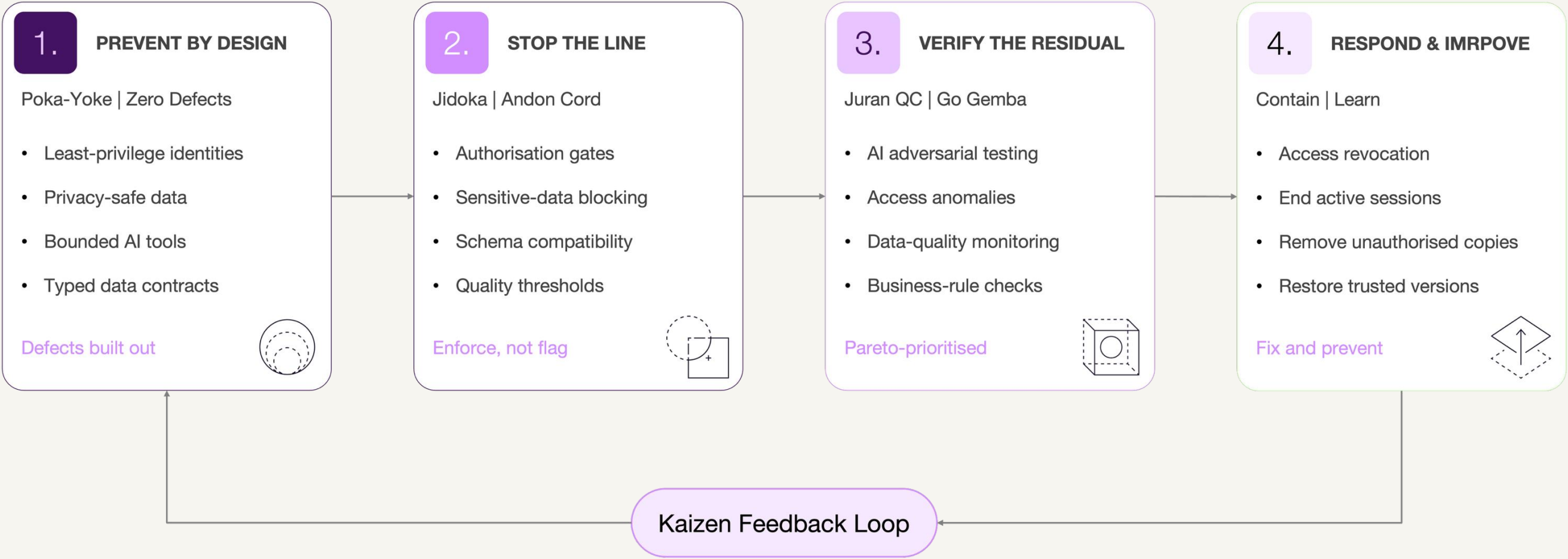


Building The Factory

How can a Japanese automotive manufacturer provide the blueprint for the software factory?

Pressure: AI writes code & finds flaws/exploits at machine speed

EXECUTION PATH (Data & Security-By-Default)



Poka-Yoke

- Any mechanism in a process that helps an operator avoid mistakes and prevent defects

Jidoka

- Intelligent automation, where machines or workers stop a process immediately upon finding a defect

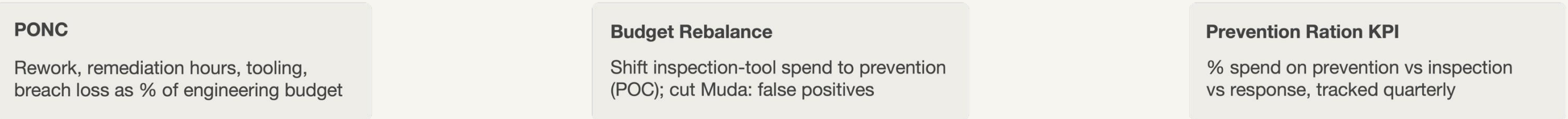
Go Gemba

- Visit the actual place where work is done and value is created

Kaizen

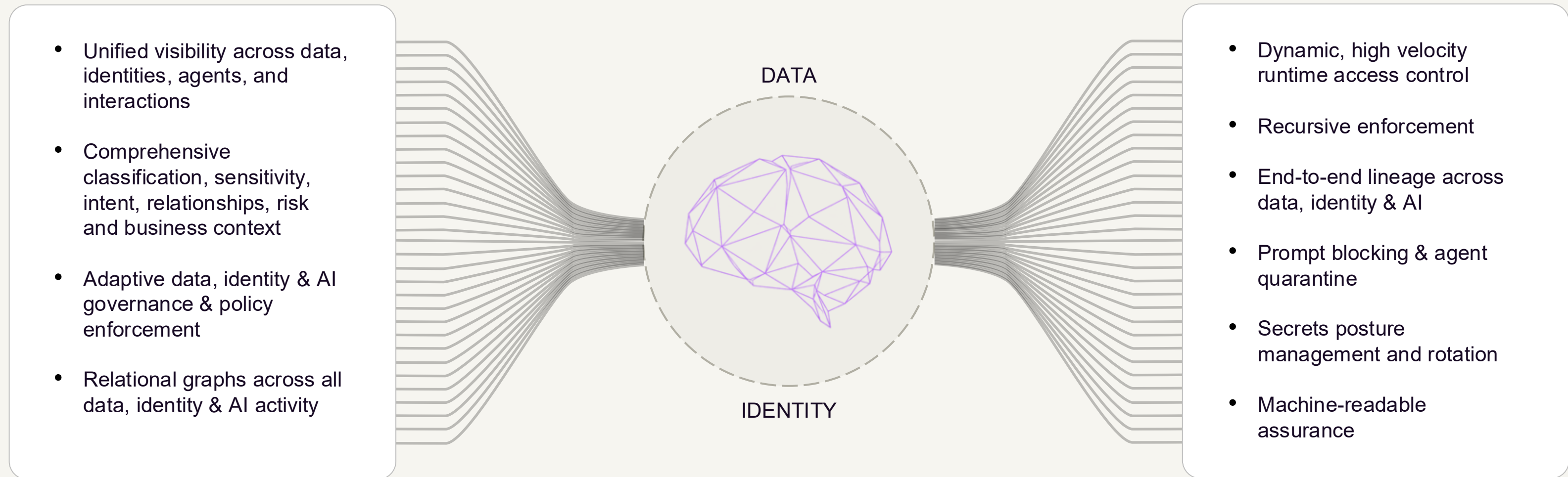
- Continuous, incremental improvements

ECONOMIC FOUNDATION (Cost of Quality)



Running The Factory

What are the first principles of the software factory?



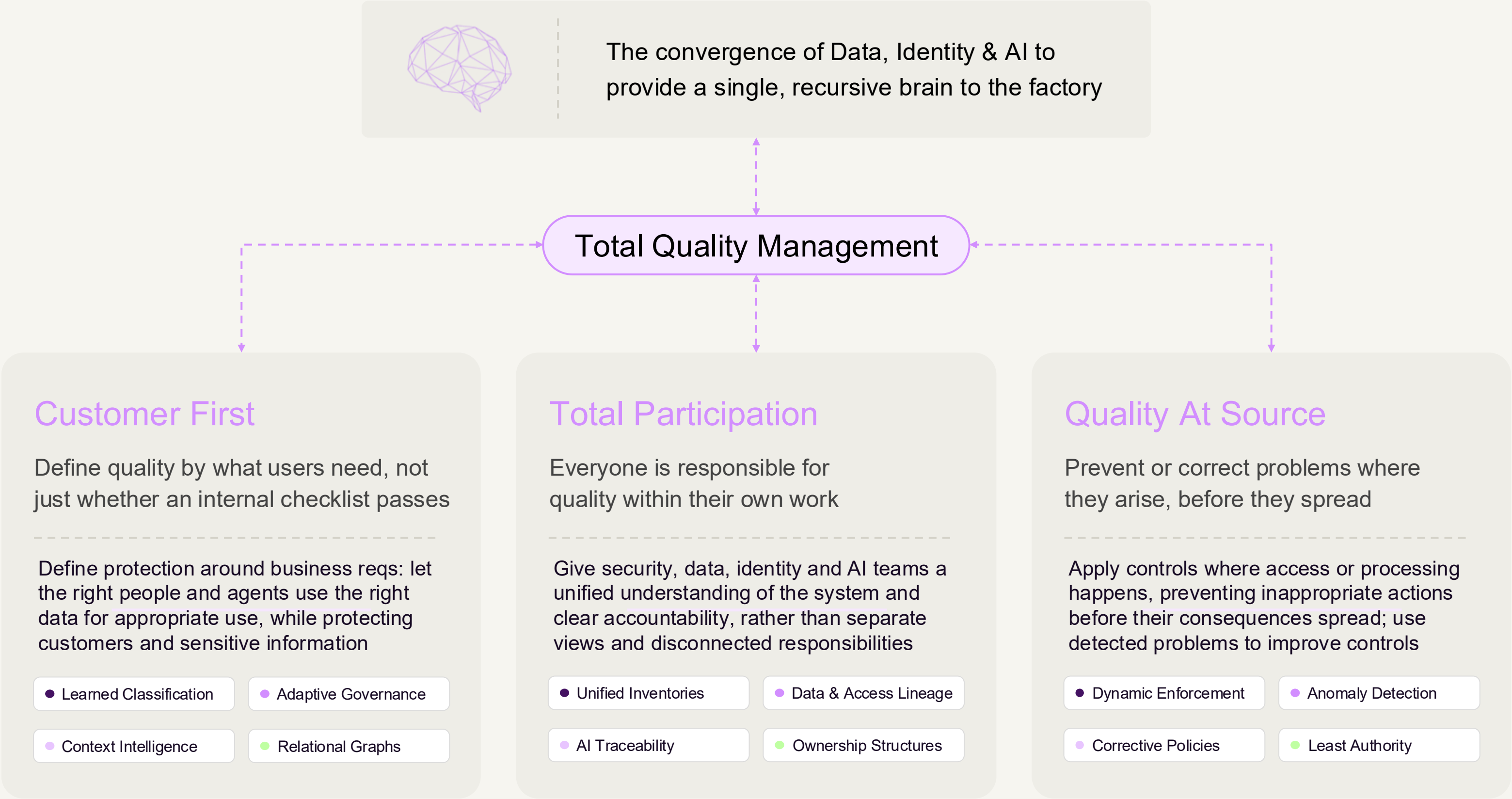
Trust Architecture

Unified Posture & Context | Dynamic Runtime Enforcement | Comprehensive Lineage & Traceability



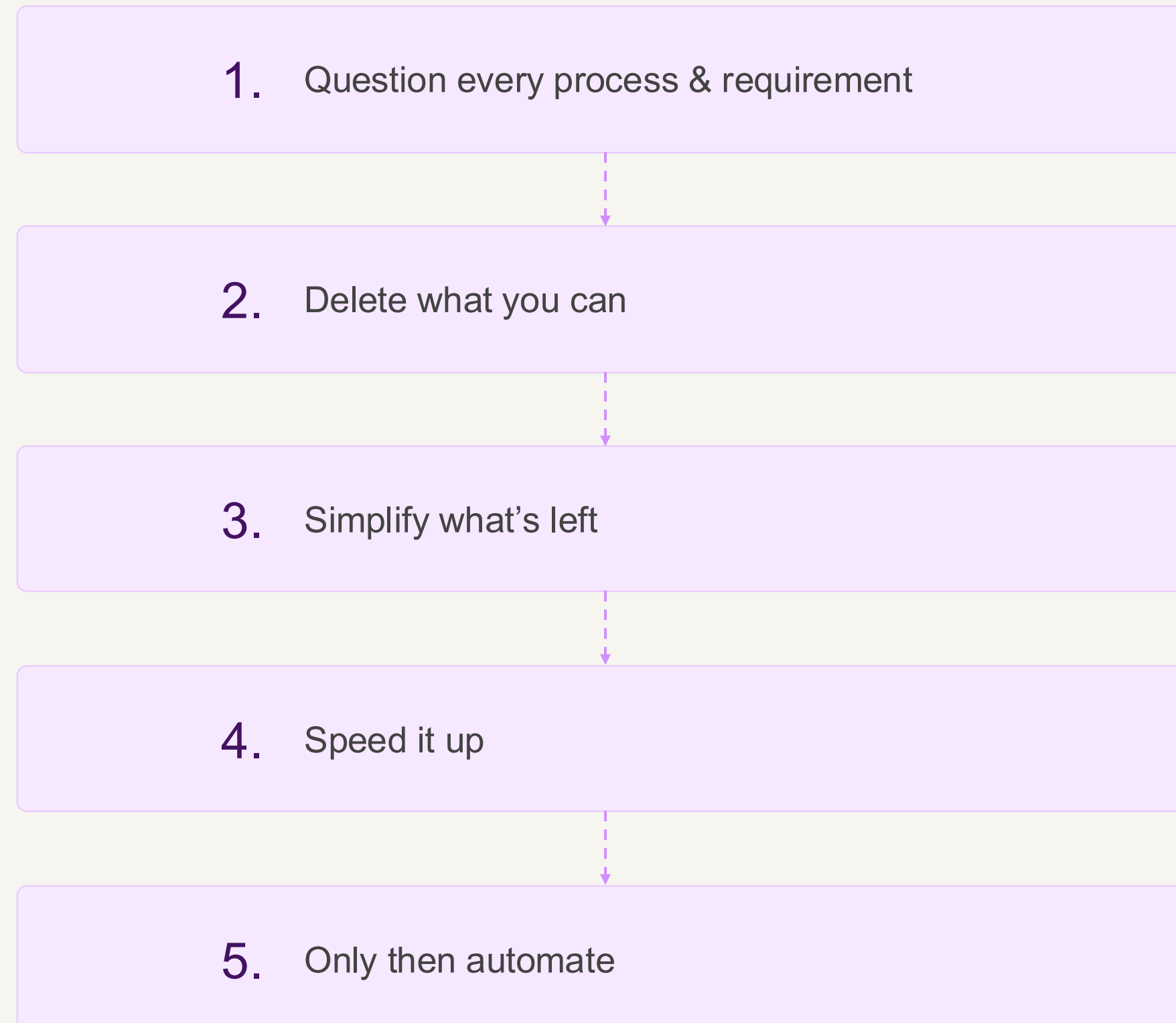
Total Quality Management

How can we secure the factory?



Rebuild & Transform

How can we begin building the factory?





Thank You

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